



Informacoes Tecnicas

Trigonometria

Material tecnico organizado pela Alta Press para consulta em projetos, compras e manutencao industrial.

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Fórmulas para auxílio no desenvolvimento de projetos e especificações de materiais:

Triângulo Retângulo

$$\text{Ângulo } A+B = 90^\circ \quad \tan A = \frac{a}{b}$$

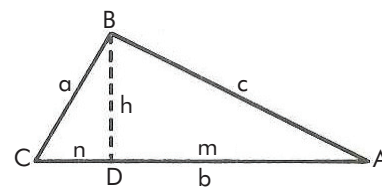
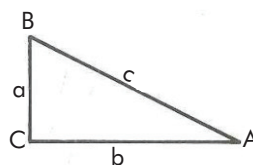
$$\sin A = \frac{a}{c}$$

$$a^2 + b^2 = c^2$$

$$\cos A = \frac{b}{c}$$

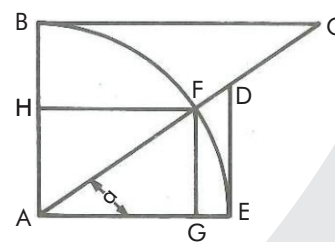
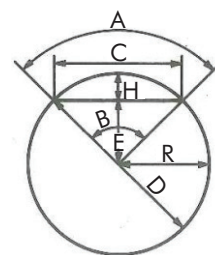
$$\text{Área} = 1/2 ab$$

Dado	Pedido	Fórmulas
A, B, a	C, b, c	$C = 180^\circ - A - B$ $b = \frac{a \sin B}{\sin A}$ $c = \frac{a \sin C}{\sin A}$
b, c, A	B, C, a	$\cos B = \frac{c^2 + a^2 - b^2}{2ac}$ $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ $a^2 = b^2 + c^2 - 2bc \cos A$ $h = c \sin A$ $m = c \cos A$ $n = b - m$ $a^2 = h^2 + n^2$
a, c, A	B, C, b	$\sin C = \sin A \times \frac{c}{a}$ $B = 180^\circ - A - C$ $b = \frac{a \sin B}{\sin A}$
a, b, c	A	$\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{b \times c}}$ $\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{b \times c}}$ $\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$ $s = 1/2 (a + b + c)$
a, b, c	m, n	$m = \frac{c^2 + b^2 - a^2}{2b}$ $n = \frac{a^2 + b^2 - c^2}{2b}$



$$\begin{aligned} \overline{AB} &= c & \overline{BC} &= a \\ \overline{AC} &= b & \overline{BD} &= h \\ \overline{AD} &= m & \overline{DC} &= n \end{aligned}$$

A, B e C são ângulos.



$$\text{Raio } AE = 1$$

$$FG = \sin a \quad BC = \cot a$$

$$AG = \cos a \quad AD = \sec a$$

$$DE = \tan a \quad AC = \csc a$$

Círculo

$$B = \text{ângulo em graus.} \quad C = 2\sqrt{2HR - H^2} = 2R \sin \frac{B}{2}$$

$$A = \text{comprimento do arco.} \quad H = R - 1/2\sqrt{4R^2 - C^2} = 2R \sin \frac{B}{4}$$

$$\text{Circunferência} = \pi D = 2\pi R \quad E = R - H$$

$$\text{Área} = \pi R^2 \quad \text{Área do Setor} = 1/2 RA$$

$$\text{Área} = \frac{\pi RB}{180} \quad \text{Área do Segmento} = 1/2 RA - 1/2 EC$$

Fórmulas Trigonométricas

$$\sin^2 a + \cos^2 a = 1 \quad \tan a = \frac{\sin a}{\cos a} \quad \sec^2 a - \tan^2 a = 1$$

$$\sin a \quad \sqrt{1 - \cos^2 a} \quad \frac{\tan a}{\sqrt{1 + \tan^2 a}} \quad \frac{1}{\sqrt{1 + \cot^2 a}}$$

$$\cos a \quad \sqrt{1 - \sin^2 a} \quad \frac{1}{\sqrt{1 + \tan^2 a}} \quad \frac{\cot a}{\sqrt{1 + \cot^2 a}}$$

$$\tan a \quad \frac{\sin a}{\sqrt{1 - \sin^2 a}} \quad \frac{\sqrt{1 - \cos^2 a}}{\cos a} \quad \frac{1}{\cot a}$$

$$\cot a \quad \frac{\sqrt{1 - \sin^2 a}}{\sin a} \quad \frac{\cos a}{\sqrt{1 - \cos^2 a}} \quad \frac{1}{\tan a}$$